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### MEDIUM TRANSPORT DEVICE, CORRUGATION UNIT, RECORDING DEVICE

#### Abstract

A medium transport device includes a corrugation unit including a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium, and a transport path through which the medium is transported. When a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position thereof in the intersecting direction are adjustable.

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## Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-029618, filed Feb. 29, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

### BACKGROUND

#### 1. Technical Field

[0002] The present disclosure relates to a medium transport device that transports a medium. Further, the present disclosure relates to a corrugation unit that imparts a corrugated shape to a medium. Further, the present disclosure relates to a recording device that performs recording on a medium.

#### 2. Related Art

[0003] A sheet discharge device disclosed in JP 2021-100874 A includes a plurality of corrugation members. The plurality of corrugation members deform a sheet into a corrugated shape in a width direction by coming into contact with the sheet, thereby enhancing a property of proceeding straight of the sheet.

[0004] Further, in the sheet discharge device disclosed in JP 2021-100874 A, some of the plurality of corrugation members are provided so as to be displaceable along the width direction, and positions that come into contact with the sheet can be changed according to a size of the sheet.

[0005] When a corrugated shape is formed in a sheet, it is required that a degree of freedom in forming the corrugated shape is further increased.

### SUMMARY

[0006] A medium transport device of the present disclosure for solving the above-described problems is a medium transport device including a corrugation unit including a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium, and a transport path through which the medium is transported, wherein when a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position in the intersecting direction are adjustable.

[0007] Additionally, a corrugation unit of the present disclosure is a corrugation unit including a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium, wherein when a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position in the intersecting direction are adjustable.

[0008] Further, a recording device of the present disclosure includes a recording unit configured to perform recording on a medium, and the above medium transport device.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a medium transport path of a printer.

[0010] FIG. 2 is a diagram illustrating the medium transport path of the printer.

[0011] FIG. 3 is a perspective view of a corrugation unit in a mounted state.

[0012] FIG. 4 is a perspective view of a mounting frame for mounting the corrugation unit.

[0013] FIG. 5 is a front view of the corrugation unit in the mounted state.

[0014] FIG. 6 is an enlarged view of a vicinity of a discharge roller pair in FIG. 5.

[0015] FIG. 7 is a cross-sectional view taken along line B-B in FIG. 5.

[0016] FIG. 8 is a cross-sectional view taken along line A-A in FIG. 5.

[0017] FIG. 9 is a side view of a corrugation member.

[0018] FIG. 10 is a diagram illustrating a corrugation formed in a medium.

[0019] FIG. 11 is a diagram illustrating a corrugation formed in a medium.

[0020] FIG. 12 is a diagram illustrating a corrugation formed in a medium.

[0021] FIG. 13 is a diagram illustrating a configuration for making a corrugation member detachable.

[0022] FIG. 14 is a diagram illustrating a configuration for switching a height direction position of a corrugation member.

[0023] FIG. 15 is a diagram illustrating a configuration for switching a height direction position of the corrugation member.

[0024] FIG. 16 is a diagram illustrating a configuration for switching a height direction position of a corrugation member.

[0025] FIG. 17 is a diagram illustrating a configuration for switching a width direction position of the corrugation member.

[0026] FIG. 18 is a diagram illustrating the configuration for switching the width direction position of the corrugation member.

[0027] FIG. 19 is a block diagram illustrating a control system of the printer.

#### DESCRIPTION OF EMBODIMENTS

[0028] Hereinafter, the present disclosure will be described in brief.

[0029] A medium transport device according to a first aspect is a medium transport device including a corrugation unit including a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium, and a transport path through which the medium is transported, wherein when a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position in the intersecting direction are adjustable.

[0030] According to the aspect, the position of the corrugation member in the width direction and the position in the intersecting direction are adjustable, thus when a corrugated shape is formed in the medium, it is possible to further increase a degree of freedom in forming the corrugated shape. This makes it possible to more appropriately transport the medium.

[0031] A second aspect is an aspect dependent on the first aspect, wherein the corrugation unit includes a supporting member that is a member for supporting the corrugation member and is movable in the width direction and the intersecting direction, and a fixing member for fixing the supporting member.

[0032] According to the aspect, the positions of the corrugation member in the width direction and the intersecting direction can be appropriately adjusted and held by the supporting member and the fixing member.

[0033] A third aspect is an aspect dependent on the second aspect, wherein the corrugation unit includes a shaft that extends along the width direction, the shaft supporting the corrugation member together with the supporting member, and the corrugation member changes a position in contact with the medium by rotating around the shaft.

[0034] According to the aspect, the corrugation member is configured to change the position in contact with the medium by rotating around the shaft, and is configured to regulate posture by the shaft and the supporting member, thus the position coming into contact with the medium of the corrugation member is stabilized. Accordingly, a corrugated shape can be formed in the medium more appropriately.

[0035] A fourth aspect is an aspect dependent on the third aspect, wherein the corrugation member and the supporting member are engaged with each other by an engagement shaft provided on one

side and an engagement groove provided on another side, and leeway is provided between the engagement shaft and the engagement groove.

[0036] According to the aspect, the leeway is provided between the engagement shaft and the engagement groove, thus it is possible to prevent the engagement shaft from applying strong force to the engagement groove when the supporting member is displaced in the intersecting direction. Thus, deformation of the supporting member or the corrugation member can be suppressed.

[0037] A fifth aspect is an aspect dependent on the first aspect, wherein the corrugation unit includes a transport roller that is a roller driven by a driving source and transports the medium.

[0038] In the configuration in which the corrugation member imparts a corrugated shape to the medium, transport resistance is generated in the medium. According to the aspect, the corrugation unit includes the transport roller that is the roller driven by the driving source and transports the medium, thus it is possible to appropriately transport the medium even when transport resistance is generated in the medium by the corrugation member.

[0039] Note that the aspect may also be dependent on any one of the above first to fourth aspects, not only on the above first aspect.

[0040] A sixth aspect is an aspect dependent on the fifth aspect, wherein the transport roller and the corrugation member overlap each other in the transport direction when viewed from the width direction.

[0041] According to the aspect, the transport roller and the corrugation member overlap each other in the transport direction when viewed from the width direction, thus the transport roller applies transport force to the medium at a position close to the corrugation member. As a result, it is possible to appropriately transport the medium even when transport resistance is generated in the medium by the corrugation member.

[0042] A seventh aspect is an aspect dependent on any one of the first to sixth aspect, wherein the corrugation member is attachable to and detachable from the corrugation unit.

[0043] According to the aspect, the corrugation member is attachable to and detachable from the corrugation unit, thus the number of corrugation members can be adjusted, and a degree of freedom in adjusting a corrugated shape to be imparted to the medium is improved.

[0044] An eighth aspect is an aspect dependent on any one of the first to sixth aspects, wherein the plurality of corrugation members include a first corrugation member coming into contact with a first surface of the medium and a second corrugation member coming into contact with a second surface of the medium opposite to the first surface.

[0045] According to the aspect, the plurality of corrugation members include the first corrugation member coming into contact with the first surface of the medium and the second corrugation member coming into contact with the second surface of the medium opposite to the first surface, thus the degree of freedom in adjusting a corrugated shape to be imparted to the medium is improved.

[0046] Note that the aspect may be dependent not only on any one of the first to sixth aspects but also on the seventh aspect.

[0047] A ninth aspect is an aspect dependent on the eighth aspect, wherein the corrugation member is attachable to and detachable from the corrugation unit.

[0048] According to the aspect, the corrugation member is attachable to and detachable from the corrugation unit, thus the number of corrugation members can be adjusted, and the degree of freedom in adjusting a corrugated shape to be imparted to the medium is further improved.

[0049] A tenth aspect is an aspect dependent on any one of the first to sixth aspects, wherein a plurality of the corrugation members include a right corrugation member positioned on a right side with respect to a center position in the width direction of the medium, and a left corrugation member positioned on a left side with respect to the center position, and the right corrugation member and the left corrugation member are provided so as to approach and separate from each other in the width direction.

[0050] According to the aspect, since the right corrugation member and the left corrugation member are provided so as to approach and separate from each other in the width direction, the right corrugation member and the left corrugation member can be disposed at positions constantly symmetrical with respect to the center position in the width direction. As a result, a corrugated shape can be appropriately imparted to the medium.

[0051] Note that the aspect may be dependent not only on the above first to sixth aspects but also on any one of the seventh to ninth aspects.

[0052] An eleventh aspect is an aspect dependent on the eighth aspect, wherein a plurality of at least one of the first corrugation members and the second corrugation members are provided along the width direction, and positions thereof in the intersecting direction change in conjunction with each other.

[0053] According to the aspect, the plurality of at least one of the first corrugation members and the second corrugation members are provided along the width direction, and the positions thereof in the intersecting direction change in conjunction with each other, thus a corrugated shape can be appropriately imparted to the medium.

[0054] Note that the aspect may be dependent not only on the above eighth aspect but also on the above ninth or tenth aspect.

[0055] A twelfth aspect is an aspect dependent on the first aspect including a frame constituting a base body of the device, wherein the corrugation unit is attachable to and detachable from the frame.

[0056] According to the aspect, the corrugation unit is attachable to and detachable from the frame, thus it is possible to reduce an installation space of the device by removing the corrugation unit when unnecessary, and usability is improved.

[0057] A corrugation unit according to a thirteenth aspect is a corrugation unit including a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium, wherein when a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position in the intersecting direction are adjustable.

[0058] According to the aspect, the position of the corrugation member in the width direction and the position in the intersecting direction are adjustable, thus when a corrugated shape is formed in the medium, it is possible to further increase the degree of freedom in forming the corrugated shape. This makes it possible to more appropriately transport the medium.

[0059] A medium transport device according to a fourteenth aspect is a medium transport device including a transport path through which a medium is transported, and includes a mounting portion from which the corrugation unit according to the thirteenth aspect is detachable.

[0060] According to the aspect, by mounting the corrugation unit to the mounting portion, operational effects of the above thirteenth aspect can be obtained. Additionally, the corrugation unit is attachable to and detachable from the mounting portion, thus it is possible to reduce the installation space of the device by removing the corrugation unit when unnecessary, and usability is improved.

[0061] A recording device according to a fifteenth aspect includes a recording unit configured to perform recording on a medium, and the medium transport device according to the first or fourteenth aspect.

[0062] According to the aspect, in the recording device including the recording unit that performs recording on the medium, operational effects of the above first or fourteenth aspect can be obtained.

[0063] Note that the medium transport device included in the aspect may be not only the above first or fourteenth aspect, but may be any one of the second to twelfth aspects.

[0064] Hereinafter, the present disclosure will be described in detail.

[0065] Hereinafter, an inkjet printer **1** that performs recording by ejecting ink, which is an example of liquid, onto a medium, such as recording paper, will be described as an example of a recording device. Hereinafter, the inkjet printer **1** will be abbreviated as printer **1**.

[0066] Note that the printer **1** can be regarded as a medium transport device **50** from the perspective of transporting the medium. In this case, the printer **1** includes the medium transport device **50** and a line head **12** which is an example of a recording unit described below. However, the entire device of the printer **1** including the line head **12** may be regarded as the medium transport device **50**.

[0067] An X-Y-Z coordinate system illustrated in each drawing is an orthogonal coordinate system, and a Y-axis direction is a medium width direction and is a device depth direction. In the embodiment, among side surfaces constituting a periphery of a device body **2**, a side surface in a +Y direction is a back surface, and a side surface in a -Y direction is a front surface.

[0068] An X-axis direction is a device width direction, and a +X direction is a left side and an -X direction is a right side as viewed by an operator of the printer **1**.

[0069] A Z-axis direction is a vertical direction, that is, a device height direction, a +Z direction is an upward direction, and a -Z direction is a downward direction.

[0070] Hereinafter, a direction in which the medium is sent may be referred to as “downstream” and an opposite direction may be referred to as “upstream”. In FIG. **1**, the medium transport path is illustrated by a dashed line. In the printer **1**, the medium is transported through the medium transport path illustrated by the dashed line.

[0071] The printer **1** includes a medium cassette **3** at a lower portion of the device main body **2** including the line head **12** described below. A reference sign P denotes the medium accommodated in the medium cassette **3**.

[0072] A pick roller **21** that sends out the accommodated medium in the -X direction is provided for the medium cassette **3**. Further, a feed roller pair **25** that feeds the medium sent out by the pick roller **21** further downstream is provided for the medium cassette **3**. Note that a plurality of medium cassettes (not illustrated) are further provided below the medium cassette **3**. Additionally, a pick roller (not illustrated) and a feed roller pair (not illustrated) are provided for each of the plurality of medium cassettes (not illustrated).

[0073] Note that, in the present specification, unless otherwise specified, it is assumed that a “roller pair” includes a drive roller driven by a power source such as a motor, and a driven roller that rotates in a driven manner in contact with the drive roller.

[0074] A reference sign T1 denotes a feeding path for the medium that is sent out from the medium cassette **3** and reaches a transport roller pair **34**. The medium sent out from the medium cassette **3** receives feed force from transport roller pairs **29** and **33** and is sent to the transport roller pair **34**. The medium receiving the feed force from the transport roller pair **34** is sent to a recording position between the line head **12** and a transport belt **53**, that is, facing the line head **12**.

[0075] The line head **12** performs recording by ejecting ink, which is an example of liquid, from a nozzle **13** provided at a nozzle surface **12a**. In the embodiment, the line head **12** is an ink ejection head in which a plurality of the nozzles **13** that eject the ink are disposed to cover an entire area in the medium width direction, and is configured as an ink ejection head that can perform recording over the entire width of the medium without moving in the width direction of the medium. However, the ink ejection head is not limited thereto, and may be a type that is mounted on a carriage and ejects ink while moving in the medium width direction.

[0076] The line head **12** according to the embodiment employs a piezo element, which is a piezoelectric element whose volume changes when a voltage is applied. A drive waveform of the piezo element is controlled so that a movement of a meniscus of the nozzle **13** can be controlled and a size and ejection speed of ejected ink droplets can be controlled.

[0077] In the embodiment, the plurality of nozzles **13** include a plurality of the nozzles **13** that eject yellow ink, a plurality of the nozzles **13** that eject magenta ink, a plurality of the nozzles **13** that

eject cyan ink, and a plurality of the nozzles **13** that eject magenta ink.

[0078] Next, the transport belt **53** is an endless belt that is wound around a first roller **54**, which is a driving roller, and a second roller **55**, which is a driven roller, and rotates when the first roller **54** is driven by a motor (not illustrated). The medium is transported to a position facing the line head **12** while being adsorbed to a belt surface of the transport belt **53**. The first roller **54**, the second roller **55**, and the transport belt **53** constitute a belt unit **52**.

[0079] The medium on which recording is performed by the line head **12** is sent to either a transport roller pair **36** or a transport roller pair **40** by a transport roller pair **35** positioned downstream of the transport belt **53**. A path switching flap (not illustrated) is provided near a downstream section of the transport roller pair **35**, and the medium receiving the feed force from the transport roller pair **35** is sent to any one of the transport roller pair **36** and the transport roller pair **40** by this path switching flap.

[0080] When double-sided recording is not performed, the medium is sent from the transport roller pair **35** toward the transport roller pair **36**.

[0081] Downstream of the transport roller pair **36**, the transport path branches into three discharge paths **T4**, **15**, and **T6**. A flap (not illustrated) that switches a transport destination of the medium is provided near a downstream section of the transport roller pair **36**.

[0082] When the discharge path **T4** is selected, the medium is discharged toward a discharge tray **8** through the discharge path **T4**. The discharge path **T4** is provided with a transport roller pair **38** and a transport roller pair **39**.

[0083] When the discharge path **T5** is selected, the medium is discharged in the +X direction through the discharge path **T5**. At this time, as long as a corrugation unit **60** to be described later is mounted, the medium is discharged to the discharge tray **4** in a state in which a corrugated shape is formed by the corrugation unit **60**. A reference sign **T7** denotes a transport path in the corrugation unit **60**. Note that the discharge path **T5** is provided with a transport roller pair **44**.

[0084] When the discharge path **T6** is selected, the medium is discharged in the +X direction through the discharge path **T6**. At this time, as illustrated in FIG. 2, as long as the corrugation unit **60** to be described later is mounted, the medium is discharged to the discharge tray **4** in a state in which a corrugated shape is formed by the corrugation unit **60**. Note that the discharge path **T6** is provided with a transport roller pair **45**.

[0085] On a left side surface of the printer **1**, a post-processing device (not illustrated) can be mounted instead of the corrugation unit **60**, and when this post-processing device is mounted, the medium is sent to the post-processing device via the discharge path **T6**. Examples of the post-processing include binding processing using a binding needle, punching processing, and the like.

[0086] When double-sided recording is performed, the medium is sent from the transport roller pair **35** to the transport roller pair **40** and enters a switchback path **T2**. Thereafter, a rotation direction of the transport roller pair **40** is switched, the medium enters a reversal path **T3**, and is sent to the transport roller pair **34** by the transport roller pairs **41**, **42**, and **43**.

[0087] A reference sign **10** denotes an ink accommodation portion serving as a liquid accommodation portion that accommodates ink before ejection. Ink ejected from the line head **12** is supplied from the ink accommodation portion **10** to the line head **12** via a tube (not illustrated). The ink accommodation portion **10** accommodates black, yellow, magenta, and cyan inks as an example.

[0088] Next, the corrugation unit **60** will be described in detail with reference to FIG. 3 and subsequent drawings.

[0089] The corrugation unit **60** forms a corrugated shape in the medium to increase rigidity thereof, suppresses curling or sagging when the medium is discharged, and improves a stacking state, and in the embodiment, is attachable to and detachable from the printer **1**.

[0090] As illustrated in FIG. 3, the corrugation unit **60** includes a base frame **61** as a base body. The base frame **61** includes side frame portions **61a** and **61b** spaced apart from each other in the Y-axis

direction, and the side frame portions **61a** and **61b** form surfaces parallel to an X-Z plane. The side frame portions **61a** and **61b** are provided with a fixed portion **61e** so as to form a surface parallel to a Y-Z plane. The fixed portion **61e** can be fixed to a mounting frame **57** by a screw (not illustrated). [0091] More specifically, the printer **1** includes the mounting frame **57** illustrated in FIG. **4**. In the mounting frame **57**, fixing portions **57a** forming a surface parallel to the Y-Z plane are formed at intervals in the Y-axis direction. A boss **57c** and a screw hole **57d** are formed at the fixing portion **57a**.

[0092] On the other hand, as illustrated in FIG. **3**, a plurality of holes **61g** are formed at the fixed portion **61e** of the corrugation unit **60** at intervals along the Z-axis direction. The boss **57c** of the mounting frame **57** can be fitted into any one of the holes **61g**. Further, a screw (not illustrated) can be fitted into the screw hole **57d** through any one of the holes **61g**. Thus, the corrugation unit **60** can be fixed to the mounting frame **57**.

[0093] Note that a hook portion **57b** is formed at the fixing portion **57a** of the mounting frame **57**. A boss **61f** is provided at the base frame **61** of the corrugation unit **60**, and the boss **61f** can be caught by the hook portion **57b**. That is, when the corrugation unit **60** is fixed to the mounting frame **57**, the corrugation unit **60** can be first hooked on the hook portion **57b** to be temporarily fixed, so that the corrugation unit **60** can be mounted with easy workability.

[0094] When the corrugation unit **60** is fixed at a plurality of height positions in the Z-axis direction as illustrated in FIGS. **1** and **2**, a plurality of sets each including the hook portion **57b**, the boss **57c**, and the screw hole **57d** are formed in the Z-axis direction.

[0095] Next, as illustrated in FIG. **3**, coupling frames **62**, **75** and **76** extending in the Y-axis direction are fixed to the side frame portions **61a** and **61b** of the base frame **61**. Thus, rigidity of the base frame **61** is enhanced.

[0096] The coupling frame **76** is provided with a roller supporting member **77**. The roller supporting member **77** is provided with a driving roller **65a** constituting a discharge roller pair **65**.

[0097] The discharge roller pair **65** is a roller pair that discharges the medium from the corrugation unit **60**, and includes the driving roller **65a** that is driven by a power source and a driven roller **65b** (see FIG. **6**) that is driven to rotate. A plurality of the discharge roller pairs **65** are provided at intervals along the medium width direction.

[0098] Next, the corrugation unit **60** includes a plurality of corrugation members **66** along the Y-axis direction, that is, the medium width direction. Note that the reference sign **66** of the corrugation member is added with a capital alphabet letter depending on an arrangement position in the medium width direction. A reference sign **66A** denotes a corrugation member positioned on an innermost side in the medium width direction. A reference sign **66B** denotes a corrugation member positioned on an outer side with respect to the corrugation member **66A** in the medium width direction. A reference sign **66C** denotes a corrugation member positioned on an outer side with respect to the corrugation member **66B** in the medium width direction.

[0099] Note that, hereinafter, when the plurality of corrugation members are not distinguished from each other, the corrugation members are collectively referred to as the corrugation members **66**.

[0100] In FIG. **5**, a reference sign CL denotes a center line in the medium width direction, and the center line CL is a center position in the medium width direction regardless of a medium size. The corrugation members **66** are disposed left and right with the center line CL as a center in the medium width direction. In particular, in FIG. **5**, the corrugation members **66** are disposed at positions that are line-symmetric with respect to the center line CL in the medium width direction.

[0101] Note that the discharge roller pairs **65** are similarly disposed at positions that are line-symmetric with the center line CL as a center in the medium width direction. Further, driven rollers **71** (see FIG. **6**) are similarly disposed at positions that are line-symmetric with the center line CL as a center in the medium width direction.

[0102] When an upper surface of the medium transported through the corrugation unit **60** is a first surface and a lower surface is a second surface, the corrugation member **66A** faces the first surface



of the medium and the corrugation members **66B** and **66C** face the second surface of the medium. [0103] In the embodiment, the plurality of corrugation members **66** have the same structure, and hereinafter, the structure of the corrugation member **66** will be described using the corrugation member **66B** illustrated in FIG. **9** as a representative example. A shaft hole **66a** and an engagement groove **66b** are formed at the corrugation member **66**. The shaft hole **66a** is a hole through which a first shaft **63** (see FIGS. **3** and **4**) or a second shaft **64** (see FIGS. **3** and **4**) passes. The corrugation member **66** is rotatable in the X-Z plane by the first shaft **63** or the second shaft **64** passing through the shaft hole **66a**.

[0104] Note that the first shaft **63** and the second shaft **64** are shafts extending along the Y-axis direction and are supported by the side frame portions **61a** and **61b**. The corrugation member **66** is movable in the medium width direction while being guided by the first shaft **63** or the second shaft **64**.

[0105] A plurality of driven rollers **69** are provided at the corrugation member **66** so as to be exposed from the corrugation member **66**. In the embodiment, the driven roller **69** is a toothed roller including a plurality of teeth along an outer periphery thereof. In the embodiment, the three driven rollers **69** are provided along a medium transport direction. The corrugation member **66** can come into contact with the medium via the plurality of driven rollers **69**.

[0106] The engagement groove **66b** is formed so as to extend in a radial direction from an axial center of the first shaft **63** or the second shaft **64**. An engagement shaft **67b** (see FIGS. **7** and **8**) provided at the supporting member **67** (see FIGS. **3** and **4**) is inserted into the engagement groove **66b**. Since the engagement groove **66b** is a long groove, the engagement shaft **67b** is inserted into the engagement groove **66b** with leeway being included.

[0107] Of the plurality of corrugation members **66**, the corrugation member **66A** is rotatable about the first shaft **63** as illustrated in FIG. **3**, and rotation thereof about the first shaft **63** is restricted by being engaged with the supporting member **67**. Further, of the plurality of corrugation members **66**, the corrugation members **66B** and **66C** are rotatable about the second shaft **64** as illustrated in FIG. **3**, and rotation thereof about the second shaft **64** is restricted by being engaged with the supporting members **67**.

[0108] As described above, the supporting member **67** regulates a position of the corrugation member **66** in the Z-axis direction, that is, a height position.

[0109] Note that, hereinafter, a position of a constituent member of the corrugation unit **60** in the Z-axis direction may be referred to as a height position. Note that the Z-axis direction is an intersecting direction intersecting a surface of the medium discharged in the +X direction by the corrugation unit **60**. The Y-axis direction, that is, the medium width direction is a direction along the surface of the medium and is a direction intersecting the +X direction.

[0110] As illustrated in FIG. **3**, the supporting member **67** according to the embodiment has a shape extending in the Z-axis direction. An elongated hole **67a** extending in the Z-axis direction is formed at the supporting member **67**.

[0111] For example, a supporting member **67** supporting the corrugation member **66A** is fixed to the coupling frame **62**. A guide hole **62d** extending in the medium width direction is formed at the coupling frame **62**. The supporting member **67** is fixed to the coupling frame **62** by a screw **68** being inserted through the elongated hole **67a** and the guide hole **62d**, and by the screw **68** being fitted into a nut (not illustrated). This fixing includes fixing in the Y-axis direction, that is, the medium width direction, and fixing in the Z-axis direction, that is, the height direction.

[0112] Similarly, the supporting member **67** supporting the corrugation members **66B** and **66C** is fixed to a standing wall portion **61c**. The standing wall portion **61c** is a part of the base frame **61** and is a portion forming a surface parallel to the Y-Z plane. A guide hole **61d** extending in the medium width direction is formed at the standing wall portion **61c**. The supporting member **67** is fixed to the standing wall portion **61c** by the screw **68** being inserted through the elongated hole **67a** and the guide hole **61d**, and by the screw **68** being fitted into a nut (not illustrated). This fixing

includes fixing in the Y-axis direction, that is, the medium width direction, and fixing in the Z-axis direction, that is, the height direction.

[0113] Note that fine irregularities are formed at a surface of the supporting member **67**, thereby suppressing a position shift of the supporting member **67** in the height direction at the time of fixing by the screw **68**.

[0114] By the plurality of corrugation members **66** described above, it is possible to form a corrugated shape along the medium width direction as illustrated in FIGS. **10**, **11**, and **12** as an example.

[0115] Note that a reference sign **71** denotes a driven roller disposed between the discharge roller pairs **65** adjacent to each other in the medium width direction. The driven roller **71** is supported by a central corrugation member **70** as illustrated in FIG. **6**. The central corrugation member **70** is fixed to a slider **72** by a screw **74**.

[0116] The slider **72** is fixed to the coupling frame **75** by a screw **73**. More specifically, a boss **75a** is formed at the coupling frame **75**. Elongated holes **72a** and **72b** elongated in the Z-axis direction are formed at the slider **72**. The elongated holes **72b** are formed on both sides of the elongated hole **72a**. The screw **73** is fitted into a screw hole (not illustrated) of the coupling frame **75** via the long hole **72a**. Further, the boss **75a** of the coupling frame **75** enters the elongated hole **72b**. With such a configuration, the slider **72** can slide in the Z-axis direction, that is, the height direction with respect to the coupling frame **75**, and can be fixed at a predetermined height position by the screw **73**.

[0117] By adjusting the position in the Z-axis direction of the slider **72**, a height direction position of the driven roller **71** can be adjusted. A position at which the driven roller **71** comes into contact with the medium can be set to be below a medium nip position by the discharge roller pair **65**. Accordingly, a corrugated shape can be formed in the medium as illustrated in FIGS. **10** to **12** by the discharge roller pair **65** and the driven roller **71**.

[0118] FIG. **10** illustrates a case where a corrugated shape is formed in plain paper **P1** as an example. On the other hand, FIG. **11** illustrates a case where a corrugated shape is formed in thick paper **P2** as an example. The thick paper **P2** is thicker and more rigid than the plain paper **P1**. When an attempt is made to form a strong corrugated shape in such a medium, the medium is damaged and strong transporting force is required, thus there is a risk that a feed failure is caused. Therefore, in the case of the thick paper **P2**, it is suitable that the corrugated shape is made relatively gentler than that in the plain paper **P1**, or the corrugated shape is not formed.

[0119] In order to make the corrugated shape relatively gentle as compared with the case of FIG. **10** or not to form the corrugated shape, a height position of the corrugation member **66A** is raised and height positions of the corrugation members **66B** and **66C** are lowered. Further, a height position of the driven roller **71** is raised. Accordingly, as illustrated in FIG. **11**, a gentle corrugated shape can be formed at the a medium **P2**.

[0120] Next, FIG. **12** illustrates a case where plain paper **P3** having a smaller size than that of the plain paper **P1** is discharged. For example, in a state where the corrugation members **66B** and **66C** are located at the positions illustrated in FIG. **10**, when a medium width direction size becomes small, there is a risk that side ends in the medium width direction cannot be appropriately supported and an appropriate corrugation cannot be formed. In FIG. **12**, two-dot chain lines indicate the positions of the corrugation members **66B** and **66C** illustrated in FIG. **10**. Therefore, by moving the corrugation members **66** in the medium width direction according to the medium width direction size, it is possible to form an appropriate corrugated shape in the medium.

[0121] As described above, the corrugation unit **60** includes the plurality of corrugation members **66** that come into contact with the medium transported in the +X direction being the transport direction and impart a corrugated shape to the medium. The position of the corrugation member **66** in the medium width direction and the position thereof in the height direction, which is an example of a direction intersecting the surface of the medium, are adjustable. Accordingly, it is possible to

further increase the degree of freedom in forming a corrugated shape, and it is possible to more appropriately transport the medium.

[0122] In addition, in the embodiment, the corrugation unit **60** includes the supporting member **67** that is movable in the medium width direction and the height direction, and the standing wall portion **61c** and the coupling frame **62** that are examples of a fixing member that fixes the supporting member **67**. With such a configuration, it is possible to appropriately adjust and hold the positions of the corrugation member **66** in the medium width direction and the height direction.

[0123] In addition, in the embodiment, the corrugation unit **60** includes the first shaft **63** and the second shaft **64** that are shafts extending along the medium width direction and support the corrugation member **66** together with the supporting member **67**. Then, the corrugation member **66** rotates about the first shaft **63** or the second shaft **64** to change the position in contact with the medium. With such a configuration, the position at which the corrugation member **66** comes into contact with the medium is stabilized, and it is possible to more appropriately form a corrugated shape in the medium.

[0124] Further, the corrugation member **66** and the supporting member **67** are engaged with each other by the engagement shaft **67b** provided at the supporting member **67** and the engagement groove **66b** provided at the corrugation member **66**. Then, the leeway is provided between the engagement shaft **67b** and the engagement groove **66b** as illustrated in FIGS. 7 and 8. Accordingly, when the supporting member **67** is displaced in the height direction, it is possible to prevent the engagement shaft **67b** from applying strong force to the engagement groove **66b**. As a result, deformation of the supporting member **67** or the corrugation member **66** can be suppressed. Note that as a modification of the embodiment, the engagement shaft **67b** may be provided at the corrugation member **66** and the engagement groove **66b** may be provided at the supporting member **67**.

[0125] In addition, the corrugation unit **60** according to the embodiment includes the driving roller **65a** as a roller driven by a driving source and a transport roller that transports the medium.

[0126] In the configuration in which the corrugation member **66** imparts a corrugated shape to the medium, transport resistance occurs in the medium, however, the corrugation unit **60** includes the driving roller **65a** driven by the driving source as described above, it is possible to appropriately transport the medium even when transport resistance is caused in the medium by the corrugation member **66**.

[0127] Further, in the embodiment, as illustrated in FIGS. 7 and 8, the driving roller **65a** and the corrugation member **66** overlap each other in the transport direction when viewed from the medium width direction. As a result, the driving roller **65a** applies transport force to the medium at a position close to the corrugation member **66**, thus the medium can be appropriately transported even when transport resistance is caused in the medium by the corrugation member **66**.

[0128] Note that as a modification of the corrugation member **66**, it is suitable to configure the corrugation member **66** to be attachable to and detachable from the corrugation unit **60**. A corrugation member **66-1** illustrated in FIG. 13 is an example of a detachable corrugation member and includes an opening portion **66f** for receiving the second shaft **64**. An opening width of the opening portion **66f** is smaller than a shaft diameter of the second shaft **64**. As an example, the corrugation member **66** is formed of a resin material, and when the shaft hole **66a** is fitted into the second shaft **64**, the opening portion **66f** is widened by elastic deformation of a corrugation member **66-1** so that the shaft hole **66a** can be fitted to the second shaft **64**. Further, when the corrugation member **66-1** is removed from the second shaft **64**, the opening portion **66f** is widened by elastic deformation of the corrugation member **66-1**, so that the corrugation member **66-1** can be removed from the second shaft **64**.

[0129] According to such a configuration, the number of corrugation members **66** can be adjusted, and the degree of freedom in adjusting a corrugated shape to be imparted to the medium is improved.

[0130] Note that when the corrugation member **66** is attached to and detached from the corrugation unit **60**, the first shaft **63** or the second shaft **64** may be made attachable to and detachable from the corrugation unit **60** instead of the configuration in which the corrugation member **66** is attachable to and detachable from the first shaft **63** or the second shaft **64**.

[0131] In addition, in the embodiment, the plurality of corrugation members **66** include the corrugation member **66A** as a first corrugation member coming into contact with the first surface of the medium and the corrugation members **66B** and **66C** as second corrugation members coming into contact with the second surface of the medium opposite the first surface. Accordingly, the degree of freedom in adjusting a corrugated shape to be imparted to the medium is improved.

[0132] In addition, in the embodiment, the printer **1** includes the mounting frame **57** constituting a base body of the device, and the corrugation unit **60** is attachable to and detachable from the mounting frame **57**. The mounting frame **57** is an example of a mounting portion from which the corrugation unit **60** is detachable. Thus, by removing the corrugation unit **60** when unnecessary, the installation space of the device can be reduced, thereby improving usability.

[0133] Note that in the above-described embodiment, the height position of the corrugation member **66** is adjusted by moving the supporting member **67** up and down, but the present disclosure is not limited thereto. For example, the corrugation member **66-2** illustrated in FIG. **14** includes a plurality of tooth portions **66e** around the second shaft **64**. A plunger **100** is provided at a position facing the tooth portions **66e**. The plunger **100** includes a main body **101** and a pin **102** that can elastically advance and retreat with respect to the main body **101**. A height direction position of the corrugation member **66-2** is regulated by the pin **102** engaging with the toothed portion **66e**. Then, as illustrated by a change between a state **ST1** and a state **ST2**, the height direction position of the corrugation member **66-2** is switched by a position at which the pin **102** meshes with the toothed portion **66e** being switched.

[0134] Needless to say, such a configuration may be applied to the first shaft **63** and the corrugation member **66** supported by the first shaft **63**. Further, such a configuration may be applied to all or only some of the plurality of corrugation members **66**.

[0135] Further, in the configuration illustrated in FIG. **15**, the height direction position of the corrugation member **66** is regulated by a cam **106**. The cam **106** is a so-called eccentric cam, and is provided at a cam shaft **105**. The cam **106** supports the corrugation member **66**. The cam shaft **105** is driven by a motor (not illustrated). When the cam **106** is rotated by rotation of the cam shaft **105**, the height direction position of the corrugation member **66** is switched as illustrated by the change between state **ST1** and state **ST2**.

[0136] Needless to say, such a configuration may be applied to the first shaft **63** and the corrugation member **66** supported by the first shaft **63**. Further, such a configuration may be applied to all or only some of the plurality of corrugation members **66**.

[0137] In addition, in the configuration illustrated in FIG. **16**, a second shaft **64-1** is used instead of the second shaft **64** described above. The second shaft **64-1** has a D-cut shape, and a shaft hole **64a1** of a corrugation member **66-3** has a shape corresponding to the D-cut shape. The second shaft **64-1** is driven by a motor (not illustrated). When the second shaft **64-1** rotates, a height direction position of the corrugation member **66-3** is switched as illustrated by a change between state **ST1** and state **ST2**. Note that the corrugation member **66-3** is slidable in an axial direction with respect to the second shaft **64-1**.

[0138] Needless to say, such a configuration may be applied to the first shaft **63** and the corrugation member **66** supported by the first shaft **63**. Further, such a configuration may be applied to all or only some of the plurality of corrugation members **66**.

[0139] Note that in the embodiments illustrated in FIGS. **15** and **16**, the height direction positions of the plurality of corrugation members **66** provided along the medium width direction can be changed in conjunction with each other. For example, height positions of the two corrugation members **66A** can be changed in conjunction with each other. Accordingly, it is possible to

appropriately impart a corrugated shape to the medium.

[0140] Note that in the configuration including the corrugation member **66A**, that is, the first corrugation member coming into contact with the first surface of the medium, and the corrugation members **66B** and **66C**, that is, the second corrugation members coming into contact with the second surface of the medium, the following aspects can be adopted. That is, it is also possible that height positions of either the first corrugation members or the second corrugation members or of both are changeable in conjunction with each other.

[0141] Next, a configuration in which a plurality of the corrugation members **66** interlock will be described with reference to FIGS. **17** to **19**.

[0142] Note that in the embodiment, the plurality of corrugation members **66** include right corrugation members (reference signs **66A1**, **66B1**, and **66C1** in FIGS. **10** to **12**) positioned on a right side (the  $-Y$  direction) of a center position in the width direction of the medium and left corrugation members (reference signs **66A2**, **66B2**, and **66C2** in FIGS. **10** to **12**) positioned on a left side (the  $+Y$  direction) of the center position.

[0143] Note that “right” and “left” in this case are for convenience and may be reversed when the viewing direction is changed.

[0144] The right corrugation member **66A1** is provided at a first rack member **121**. The left corrugation member **66A2** is provided at a second rack member **122**. A rack portion **121a** is formed at the first rack member **121**, and a rack portion **122a** is formed at the second rack member **122**. A pinion **117** meshes with the rack portion **121a** and the rack portion **122a** to constitute a rack-and-pinion mechanism.

[0145] The right corrugation members **66B1** and **66C1** are provided at a third rack member **123**. The left corrugation members **66B2** and **66C2** are provided at a fourth rack member **124**. Rack portions (not illustrated) are formed at the third rack member **123** and the fourth rack member **124**, and a pinion **118** meshes with these rack portions to constitute a rack-and-pinion mechanism.

[0146] A pulley **114** is integrally provided at the pinion **117**. Further, a shaft **111** is provided with a pulley **113**, and an endless belt **109** is wound around the pulley **113** and the pulley **114**.

[0147] Similarly, the pinion **118** is integrally provided with a pulley **116**. Further, the shaft **111** is provided with a pulley **115**, and an endless belt **110** is wound around the pulley **115** and the pulley **116**.

[0148] The shaft **111** is rotatable and is provided with a toothed gear **112**. Power of a moving motor **89** (see FIG. **19**) is transmitted to the toothed gear **112**, and when the shaft **111** rotates, the endless belts **109** and **110** are rotated.

[0149] For example, when the shaft **111** rotates in a clockwise direction in FIG. **18**, the first rack member **121**, that is, the right corrugation member **66A1**, and the third rack member **123**, that is, the right corrugation members **66B1** and **66C1**, move in the  $+Y$  direction. Further, when the shaft **111** rotates in the clockwise direction in FIG. **18**, the second rack member **122**, that is, the left corrugation member **66A2**, and the fourth rack member **124**, that is, the left corrugation members **66B2** and **66C2**, move in the  $-Y$  direction.

[0150] Further, when the shaft **111** rotates in a counterclockwise direction in FIG. **18**, the first rack member **121**, that is, the right corrugation member **66A1**, and the third rack member **123**, that is, the right corrugation members **66B1** and **66C1**, move in the  $-Y$  direction. Further, when the shaft **111** rotates in the counterclockwise direction in FIG. **18**, the second rack member **122**, that is, the left corrugation member **66A2**, and the fourth rack member **124**, that is, the left corrugation members **66B2** and **66C2**, move in the  $+Y$  direction.

[0151] As described above, the right corrugation members (denoted by the reference signs **66A1**, **66B1**, and **66C1** in FIGS. **10** to **12**) and the left corrugation members (denoted by the reference signs **66A2**, **66B2**, and **66C2** in FIGS. **10** to **12**) are provided so as to approach and separate from each other in the medium width direction.

[0152] As a result, the right corrugation members and the left corrugation members are disposed at

positions that are constantly symmetrical with respect to the center position in the medium width direction, and it is possible to appropriately impart a corrugated shape to the medium.

[0153] Note that in the embodiment, the height position of the corrugation member **66** can be adjusted by rotating the first shaft **63** and the second shaft **64** as in the embodiment illustrated in FIG. **16**. In FIG. **19**, a first shaft driving motor **90** is a driving source of the first shaft **63**, and a second shaft driving motor **91** is a driving source of the second shaft **64**. The first shaft driving motor **90** rotates the first shaft **63** via a power transmission unit (not illustrated). The second shaft driving motor **91** rotates the second shaft **64** via a power transmission unit (not illustrated). Note that in the configuration in which the first shaft **63** and the second shaft **64** are rotated, it is suitable that a worm gear mechanism or the like is interposed in the above power transmission unit so that the height position of the corrugation member **66** is not varied by external force received from the medium.

[0154] As described above, the position of the corrugation member **66** in the medium width direction and the position thereof in a direction intersecting the surface of the medium can be adjusted by the movement unit. The movement unit can include the width direction movement unit for adjusting the position of the corrugation member **66** in the medium width direction and the intersecting direction movement unit for adjusting the position in a direction intersecting the surface of the medium.

[0155] Although the width direction movement unit includes the rack-and-pinion mechanism in the above-described example, for example, the corrugation member **66** may be directly moved in the medium width direction by an endless belt, a ball screw, or the like. In addition, for example, the corrugation member **66** may be moved in the medium width direction by a user rotating the shaft **111** without depending on the power of the motor.

[0156] Additionally, although the intersecting direction movement unit is configured to rotate the first shaft **63** and the second shaft **64** in the above-described example, the cam mechanism as described with reference to FIG. **15** may be used to move the corrugation member **66** in a direction intersecting the surface of the medium. In addition, the intersecting direction movement unit may be configured such that, for example, the user rotates the first shaft **63** and the second shaft **64** without depending on the power of the motor.

[0157] Here, a control unit **80** will be described with reference to FIG. **19**.

[0158] The control unit **80** performs various controls including recording control in the printer **1**. Note that FIG. **19** only illustrates constituent elements necessary for describing the embodiment, and does not illustrate other constituent elements.

[0159] The control unit **80** controls a discharge motor **88**, the moving motor **89**, the first shaft driving motor **90**, and the second shaft driving motor **91**. Each of the motors is a DC motor as an example. Each of the motors is provided with rotary encoders (not illustrated), and the control unit **80** can detect a rotation direction, rotation amount, and rotation speed of each of the motors using these rotary encoders. That is, the control unit **80** can detect a drive direction, a drive amount, and a drive speed of each drive object.

[0160] The discharge motor **88** is a driving source of the driving roller **65a** constituting the discharge roller pair **65**. The discharge motor **88** may be provided at a main body of the printer **1**, or may be provided at the corrugation unit **60**. When the discharge motor **88** is provided at the main body of the printer **1**, the corrugation unit **60** obtains power of the discharge motor **88** from the main body of the printer **1** via a power coupling unit (not illustrated). In addition, when the discharge motor **88** is provided at the corrugation unit **60**, the corrugation unit **60** obtains electric power from the main body of the printer **1** via a coupling portion (not illustrated).

[0161] Similarly, the first shaft driving motor **90** and the second shaft driving motor **91** may be provided at the main body of the printer **1** or may be provided at the corrugation unit **60**.

[0162] The control unit **80** includes a CPU **81** that executes a computer program, in other words, software, a volatile memory **82**, and a non-volatile memory **83**. The CPU **81** performs various

calculations required to execute a program **84** stored in the non-volatile memory **83**. The volatile memory **82** is used as a temporary data storage area. The non-volatile memory **83** stores the program **84** and control parameters **85** required to execute the program **84**. Various processes to be described later are realized by the control unit **80** executing the program **84**.

[0163] The control unit **80** drives each motor based on a medium type and medium size information included in recorded data, and adjusts the width direction position and the height direction position of the corrugation member **66** as described with reference to FIGS. **10** to **12**.

[0164] Note that when strength of a corrugation is excessive for a thickness of the medium, a load of transport by the discharge roller pair **65** increases. Therefore, when a driving current value of the discharge motor **88** exceeds a predetermined threshold value, the control unit **80** can also control each motor so as to weaken the strength of the corrugation. For example, when the drive current value of discharge motor **88** exceeds the predetermined threshold value in the state illustrated in FIG. **10**, a position of corrugation member **66** can be changed as illustrated in FIG. **11**.

[0165] The present disclosure is not limited to the embodiments and modifications described above, and it is obvious that various modifications are possible within the scope of the disclosure described in the claims, and these are also included in the scope of the present disclosure.

## Claims

1. A medium transport device, comprising: a corrugation unit including a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium; and a transport path through which the medium is transported, wherein when a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position in the intersecting direction are adjustable.
2. The medium transport device according to claim 1, wherein the corrugation unit includes a supporting member that is a member for supporting the corrugation member and is movable in the width direction and the intersecting direction and a fixing member for fixing the supporting member.
3. The medium transport device according to claim 2, wherein the corrugation unit includes a shaft that extends along the width direction, the shaft supporting the corrugation member together with the supporting member, and the corrugation member changes a position in contact with the medium by rotating around the shaft.
4. The medium transport device according to claim 3, wherein the corrugation member and the supporting member are engaged with each other by an engagement shaft provided on one side and an engagement groove provided on another side, and leeway is provided between the engagement shaft and the engagement groove.
5. The medium transport device according to claim 1, wherein the corrugation unit includes a transport roller that is a roller driven by a driving source and transports the medium.
6. The medium transport device according to claim 5, wherein the transport roller and the corrugation member overlap each other in the transport direction when viewed from the width direction.
7. The medium transport device according to claim 1, wherein the corrugation member is attachable to and detachable from the corrugation unit.
8. The medium transport device according to claim 1, wherein a plurality of the corrugation members include a first corrugation member coming into contact with a first surface of the medium and a second corrugation member coming into contact with a second surface of the medium opposite to the first surface.
9. The medium transport device according to claim 8, wherein the corrugation member is

attachable to and detachable from the corrugation unit.

**10.** The medium transport device according to claim 1, wherein a plurality of the corrugation members include a right corrugation member positioned on a right side with respect to a center position in the width direction of the medium and a left corrugation member positioned on a left side with respect to the center position, and the right corrugation member and the left corrugation member are provided so as to approach and separate from each other in the width direction.

**11.** The medium transport device according to claim 8, wherein a plurality of at least one of the first corrugation members and the second corrugation members are provided along the width direction, and have positions changing in the intersecting direction in conjunction with each other.

**12.** The medium transport device according to claim 1, comprising a frame included in a base body of the device, wherein the corrugation unit is attachable to and detachable from the frame.

**13.** A corrugation unit comprising a plurality of corrugation members that come into contact with a medium transported in a transport direction and impart a corrugated shape to the medium, wherein when a direction along a surface of the medium and intersecting the transport direction is a width direction, and a direction intersecting the surface of the medium is an intersecting direction, a position of the corrugation member in the width direction and a position in the intersecting direction are adjustable.

**14.** A medium transport device including a transport path through which a medium is transported, the medium transport device comprising a mounting portion to which the corrugation unit according to claim 13 is detachably attached.

**15.** A recording device, comprising: a recording unit configured to perform recording on a medium; and the medium transport device according to claim 1.

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